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RESEARCH ARTICLE

Power to the People! Assigning German Popular Music Genres Based on Listeners' Perspective Using a Genre Tree Approach

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Abstract

Despite the popularity of playlists on streaming platforms, music genres still remain a primary organizational tool among listeners, stakeholders, and researchers. Yet both 'correct' genre labels for individual popular music phenomena and structure of genre taxonomies are often contested. Nevertheless, assumed assignments often form the basis of music genre recognition models in MIR research as well as musical taste and recommender systems research, raising questions about reliability, validity, and comparability of results. By contrast, popular music studies have emphasized the listener-dependent nature of genre assignments. In other words: Genre assignment may be considered a matter of majority voting within a certain community. So why not make a virtue of necessity? We constructed a hierarchical tree from MusicBrainz folksonomy genre votes for 167,762 tracks listened to in Germany across the past 50 years. The resulting Community Music Genre Tree (COMGET) comprises 234 genres—usefully granular but probably too detailed for many research and applied tasks. We address this by an additional tree-folding algorithm that aggregates genres to the desired level of detail for any research problem. For automatic genre assignment of new tracks, we developed classifiers at multiple granularity levels using music features and meta-data drawn from online databases. Results indicate that the approach is able to assign genres that correspond to listeners' perspective, in a transparent, reproducible, and intersubjectively comprehensible way.

Keywords: music genre recognition, classification, network analysis, popular music in Germany

1. Introduction

Although playlist-driven listening has grown on streaming platforms, genre labels remain the primary way people and institutions organize popular music: Listeners search and browse by genre; playlist editors and DJs assemble genre playlists; A&R, marketing and label teams position and promote artists using genre tags; and music journalists and researchers rely on genre labels to describe and compare popular music (Bonini and Magaudda, 2024; Frith, 1996). Some scholars argue that classificatory practices have weakened since the late twentieth century (DiMaggio, 1987) and that certain genre boundaries are unstable (Silver et al., 2016), but empirical evidence shows that listening and social signaling remain organized around genres (Airoldi et al., 2016; Lonsdale, 2021). On streaming platforms, genre is embedded in metadata and pipeline logic—used as search filters, playlist seeds, and features in recommender systems—which means platform infrastructures amplify genre's practical significance for discovery and market positioning (Valverde and Hesmondhalgh, 2025). For researchers, genre labels serve both theoretical and methodological roles: they are variables in preference and reception studies (Lepa et al., 2020), inputs to recommender-system research (Porcaro et al., 2024), and useful aggregators for measuring perceived diversity and mitigating hubness in high-dimensional audio and metadata feature spaces (Flexer et al., 2018; Flexer and Grill, 2016).

However, 'correct' genre labels for individual popular music phenomena (i.e., tracks, artists, releases, scenes, market, playlists, labels, performances etc.) are frequently contested (van Venrooij and Schmutz, 2018). Different actors attach different labels for different reasons: artists may self-label to narrate their

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musical biography; publishers and distributors tag releases to support marketing and placement; critics and musicologists apply normative or historical taxonomies; and listeners use genre terms to organize their listening repertoires and to signal identity (Frith, 1996; Negus, 1999). Not only do chosen labels diverge, but the taxonomies they derive from differ as well. Commercial platform taxonomies illustrate these structural differences: The metadata platform *Gracenote* uses a compact set of 25 genre tags (Jiang et al., 2024), whereas *Spotify* exposes many thousands of genre tags. Platforms also differ in the level of abstraction at which they apply genre (track, release, or artist) and in assignment strategy (single primary genre versus multi-label tag sets).

These disagreements pose concrete methodological challenges for empirical researchers across MIR and computational musicology. Problems include inconsistent genre labels that degrade generalization of supervised Machine-Learning training (Bogdanov et al., 2019), survey items that ask about genre in ways listeners interpret differently (Lange et al., 2025), and cross-study noncomparability that undermines replication and meta-analysis (Sturm and Flexer, 2023). Faced with these constraints, researchers often pragmatically adopt available commercial or open taxonomies, without systematically engaging with the source of labels or underlying theoretical assumptions (Green et al., 2024). To address this gap, we present a reproducible procedure to derive a scalable and transparently constructed genre taxonomy for a given corpus by analyzing the corresponding genre discourse in a folksonomy. Paired with automated music-genre recognition, this approach can (1) surface empirical regularities and disagreements in label use that inform validity debates (Sturm and Flexer, 2023), and (2) provide a shared, probabilistic 'currency' for cross-disciplinary quantitative work—so that classifiers, preference studies, and sociological analyses operate from a common, well-documented mapping between recordings and meta-genres.

1.1 Musical Genres as hierarchical discourse formation

Musical genres are organized hierarchically: Pachet et al. Pachet et al. (2000) identified genealogical links (e.g., *rock* as parent of *alternative rock*), geographical links (*hip hop* as supergenre of *German hip hop*), and historical-period links (*pop* as supergenre of *80s pop*) in existing taxonomies. As Brackett (Brackett, 2016, p. 8) argues, hierarchy depth is not limited to two levels but can extend virtually indefinitely—yet consensus on which labels to select and how to relate them remains contested. Expert-constructed taxonomies show limited agreement (Pachet et al., 2000), and expert classifications often diverge from listener community norms(Sordo, 2008). Adopting the power-critical perspective of cultural studies (du Gay et al., 2013), we interpret genre discourse through Foucault's Foucault (2013) concept of discourse formation, in which industry, listeners, and artists exert unequal influence over which labels circulate and stabilize. Given that emerging genres are primarily defined by listeners (Lena and Peterson, 2008), genre assignment can be understood as implicit majority voting within a community of listeners.

We therefore ground our propsed method in the genre assignments of a large listener community captured in an online folksonomy. Platforms such as *RateYourMusic*, *Discogs* , or *MusicBrainz* provide crowdsourced 'truth' reflected in aggregate tagging behavior. MusicBrainz seems particularly suitable for our endevour: it maintains a detailed whitelist of over 2,000 genre tags without imposing an a priori hierarchy, and its democratic voting system allows users to assign and vote on tags at multiple levels (track, release, artist), enabling nuanced analysis of consensus and disagreement. Although we use MusicBrainz as our primary data source, external validation against other folksonomies (i.e., Discogs) remains necessary to judge generalizability (Sturm and Flexer, 2023). Schreiber Schreiber (2015, 2016) demonstrated that hierarchical genre relationships can be inferred from tag co-occurrence patterns on digital platforms. Crucially, however, genre assignment conventions vary across listener communities in different countries and time periods (Pichl et al., 2017; Nie, 2022). As a case study, we anchor our exemplary analysis in popular music tracks listened to in Germany over the past 50 years to ensure the resulting taxonomy reflects a coherent discursive context.

For the use of hierarchical genre taxonomies in empirical research, three pragmatic issues arise. First, balanced genre distributions improve statistical power and classifier performance Japkowicz and Stephen (2002). However, subgenres are typically assigned to fewer tracks than their parent categories Bogdanov et al. (2019), creating inherent imbalance that must be considered when constructing or aggregating within the taxonomy. Second, the optimal number of genres depends on the research question: surveys of musical taste may require moderate to high resolution (Lange et al., 2025, e.g., 24 genres), robust machine learning models often perform better with fewer, broader categories (Genussov and Cohen, 2010), and fine-grained subcultural analyses may demand more genres (Caparrini et al., 2020). At the same time, preserving consistency with the overall tree structure is desired. This demands a scalable taxonomy that supports multiple resolutions derived from a single underlying hierarchy. Third, applying such taxonomies to large corpora in computational contexts requires an automated classifier that can assign genres in an efficient, consistent and transparent way, serving as a neutral arbiter when human dissent over labels is expected.

1.2 Towards an automated classifier for popular music studies

Music genre recognition (MGR) is a core task in Music Information Retrieval. Green et al. Green et al. (2024) provide a systematic state-of-the-art review and critique how genre theory from the social sciences and humanities is often overlooked when selecting features (for a systematic review on genre theories see Merlini (2020)). Commonly, researchers in MIR and music psychology often assume that genre can be inferred from audio features alone (Tzanetakis and Cook, 2002; Rentfrow et al., 2011). This conflicts with sociological genre theories that describe genres as "systems of orientations, expectations and conventions that circulate between industry, text and subject" (Neale, 1980, p. 19), emphasizing extra-musical context alongside sound. Fabbri's Fabbri (1982) genre theory offers a synthesis: musical style (instrumentation, rhythm, harmony, timbre) and release milieu (subcultural context, artist positioning, market placement) together inform how listeners assign genres. Modern MIR tools can extract style features from audio signals (Bogdanov et al., 2013), and open music databases provide access to distributional metadata (label, chart performance, platform placement).

Lyrics occupy an intermediate position, conveying both emotional expression as stylistic content (Brand et al., 2019) and contextual information through language codes and topics that signal subcultural affiliation (Carbone et al., 2024). A classifier might need to pay special attention to the *mainstream pop* or *pop* metagenre that Fabbri Fabbri (1982) describes as encompassing eclectic, commercially optimized works that resist association with a specific milieu. In addition, artists from former niche scenes are often retrospectively relabeled as *pop* or *mainstream* after achieving commercial success (Holt, 2007, p. 20). Therefore, popularity itself seems to be a relevant feature dimension for classification. We thus propose a feature space spanning four dimensions—musical style, release context, lyrical content, and popularity—to operationalize the multi-dimensional constitution of genre.

Drawing on the theoretical and methodological foundations outlined above, we pose three research questions:

RQ1 How can we construct an empirically grounded and balanced hierarchical genre tree for popular music in Germany?

RQ2 How can we aggregate to a scalable number of supergenres from the inferred tree suitable for machine learning and statistical applications?

RQ3 Can an automatic classifier predict genres at different levels of detail with robust performance using stylistic, distributional, lyrical, and popularity features?

In the following sections, we will: (2) introduce a method to create a balanced hierarchical genre taxonomy based on genre co-occurrences in folksonomy tags, (3) describe a folding algorithm to reduce the number of genres for various tasks, (4) present automatic classifiers for different levels of granularity, and (5) discuss our findings and summarize the results.

2. RQ1: A Balanced Hierarchical Genre Tree

2.1 Data

We grounded the analysis on a curated corpus of 254,334 popular tracks listened to in Germany over the past 50 years (POPTRAG). The corpus comprises for sources: 215,702 tracks from all available weekly *MediaControl* single and album charts (18 Nov 1977–21 Nov 2024), which we scraped and linked to *Spotify* identifiers (linking success: 75% for distinct singles and albums); 26,949 *Spotify* recommendations produced for 126 German-market genre seeds (retrieved 29 Nov 2024; 5 iterations $\times$ 50 recommendations per seed; median recommendations per seed = 325, IQR 25, range 49–343); 8,801 tracks from 100 featured *Spotify* playlists for the German market (median 90 tracks per playlist, IQR 42, range 28–750; retrieved 29 Nov 2024); and 11,994 entries from *Germany's Daily Top 200 Spotify* charts (01 Jan 2017–30 Nov 2024). We removed likely non-musical snippets by excluding tracks shorter than 30 seconds (853 tracks) and retained only unique combinations of track title and principal artist, preferring earlier release dates when duplicates occurred

To derive usable genre labels we then filtered tags conservatively: we discarded tracks lacking genre tags or carrying only non-musical labels and removed tags absent from the *MusicBrainz* whitelist (snapshot 16 Jun 2025). Additionally, we required a tag to appear across at least 50 distinct artists to ensure community support for the genre label. Finally, we only retained tracks with at least two tag votes to reduce single-opinion noise. Non-musical tags excluded at this stage included: *non-music*, *interview*, *comedy*, *nature sounds*, *children's music*, *spoken word*, and *standup comedy*. These criteria removed 86,572 tracks (34%), yielding a *MusicBrainz*-tagged subset of 167,762 tracks annotated with 234 distinct music genre tags. Track release years spanned 1924–2024 (median 2012, IQR 16 years). When track-level tags were missing we fell back to release tags and, if necessary, to artist tags; tag origin therefore comprised track tags for 90,470 tracks (54%), release tags for 5,915 tracks (4%), and artist tags for 71,377 tracks (43%). Users assigned between 1 and 22 different tags per track (median 4, IQR 3) and cast between 2 and 135 votes per track (median 6, IQR 7). This multi-source, vote-weighted corpus provided a broad, listener-centered basis for inferring genre co-occurrences.

2.2 Methods: From Co-Occurrences to Taxonomy

We built a hierarchical genre taxonomy from MusicBrainz tag co-occurrences. For that we adapted Schreiber (2015) to make every transformation explicit and reproducible. First, we constructed a weighted co-occurrence network: nodes represented genre labels and a directed edge $e_{A\rightarrow B}$ quantified how strongly genre A predicted genre B on the same track. For each ordered pair (A, B) we computed a vote-weighted conditional frequency

$$e_{A\rightarrow B} = P(B \mid A) \times \frac{1}{|T_{A,B}|} \sum_{t \in T_{A,B}} \frac{v_{t,B}}{v_{t,\text{total}}} \tag{1}$$

where $T_{A,B}$ denotes tracks tagged A and B, $v_{t,B}$ the votes for B on track t, and $v_{t,\text{total}}$ the total votes on t. This formula produced asymmetric edges ($e_{A\rightarrow B} \neq e_{B\rightarrow A}$) that combined directional co-occurrence with community vote strength. For illustration, suppose 1,000 tracks carried *metal*; 600 of those also carried *rock*, so $P(rock|metal) = 0.6$. If the average vote share for *rock* on tracks with both *rock* and *metal* equalled 0.5, then $e_{metal\rightarrow rock} = 0.6.5 = 0.3$. Conversely, 6,000 tracks carried *rock*, 600 of those carried *metal*, so $P(metal|rock) = 0.1$; with an average *metal* vote share of 0.7 on *rock* tracks, $e_{rock\rightarrow metal} = 0.07$. These raw weights express both prevalence and tag confidence.

Second, we resolved reciprocal links to obtain a uni-directional scaffold. For each pair we subtracted the weaker directional signal from the stronger and kept only the positive residual:

$$e'_{A\rightarrow B} = \left(e_{A\rightarrow B} - e_{B\rightarrow A}\right)_+ \tag{2}$$
$$e'_{B\rightarrow A} = \left(e_{B\rightarrow A} - e_{A\rightarrow B}\right)_+ \tag{3}$$

In the example this yields $e'_{metal\rightarrow rock} = 0.6 - 0.07 = 0.53$ and $e'_{rock\rightarrow metal} = 0$, making *metal* → *rock* a clear candidate parenthood link.

Third, we enforced a strict tree by assigning each node the single outgoing edge with the largest e' weight as its parent; all other outgoing edges were discarded. Components without parents we attached to the artificial root *POPULAR MUSIC* to guarantee a connected hierarchy. In the example, unless *metal* had a stronger e' to another genre, it would be attached to *rock*; if *rock* had no parent, it would be attached to *POPULAR MUSIC*.

Finally, we balanced the tree across levels. We estimated each genre's size as the sum of its assignment probabilities across tracks, computed the Gini coefficient within each level, and defined overall imbalance as the weighted mean of level Ginis. We then searched for the least-imbalanced tree by iteratively removing the weakest parenthood edge, reattaching the orphan to *POPULAR MUSIC*, and recomputing imbalance; we selected the iteration with minimal overall Gini.

2.3 Results: Community Music Genre Tree (COMGET)

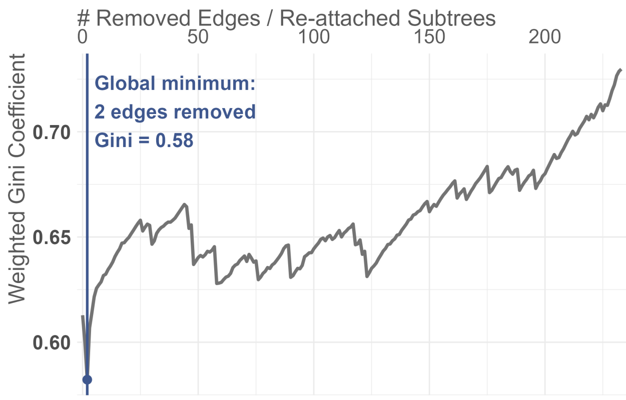

Figure 1: Weighted Gini coefficient across iterations of cutting weak genre connections and connecting resulting dangling trees to artificial root *POPULAR MUSIC*.

Level	# nodes	# tracks	Gini
1	3	total: 47,846 (29%) 12,472; 16,401; and 18,973	0.136
2	119	total: 90,381 (54%) median: 263 IQR: 789 range:10.4 to 5832	0.643
3	110	total: 29,343 (17%) median: 156 IQR: 259 range:14.7 to 1496	0.531
4	2	total: 192 (0.1%) 55.6 and 136	0.420

Table 1: Structure of COMGET at different levels of the hierarchy (distance from root of tree); number of track equals sum of genre tag assignment probabilities in POPTRAG.

The initial POPTRAG genre co occurrence network contains 234 genre nodes. The network forms a single connected component and exhibits a density of 45.14% (observed links / potential links), indicating relatively high inter tag connectivity. By median, a genre co occurred with 97 different other genres (IQR 63.5; range 3 for *waltz* to 229 for *pop*), which shows large differences between niche and ubiquitous tags. After removing weaker reciprocal directions, each genre receives on average 29.5 incoming candidate subgenre links (IQR 32; min 0; max 228 for *poprock*), and has on average 55 outgoing candidate supergenre links (IQR 32; min 0 for *rock*; max 105 for *dub*). After selecting the single strongest parent per node and running the balance optimization (see figure 1, *pop* and *hip hop* were reattached to *POPULAR MUSIC*. *pop* was a subgenre of *rock* before. *hip hop* was a subgenre of *pop* before. The three weakest links that remain (exemplary) are: *electronic* to *pop* with 0.22, *classical* to *pop* with 0.27, and *jazz* to *pop* with 0.29.), the tree becomes highly leaf heavy: 89.36% of nodes are leaves. *rock* emerges as the largest supergenre with 63 direct subgenres, and only *rock*, *pop* and *hip hop* remain directly connected to the artificial root *POPULAR MUSIC*, forming three broad meta categories that subsume the remaining tags. Please see the digital appendix for an interactive plot to explore COMGET. The tree exhibits 4 hierarchy levels (excluding artificial root genre) (see table 1 for details on the tree structure).

3. RQ2: Scalable Granularity

3.1 Methods: Folding Algorithm

The goal of this next step was to fold COMGET to a scalable number of supergenres suitable for a variety of research questions and statistical applications. The result should meet three requirements: First, like the original tree, the folded tree should be well-balanced. Second, the tree should comprise as many super genres as desired (e.g., a specific application could ask for 10 to 15 genres and the algorithm should select a solution with a suitable number). Third, each genre should have a sufficient amount of tracks associated with it to allow for machine learning tasks as robust statistics. We designed a simple greedy algorithm that can be applied to COMGET and ensures that all three requirements are met: The algorithm iteratively merges the least populous genres that are furthest from the root with their parent genres until all genres meet the minimum song count requirement. That means that subgenres that do not meet a minimum song count requirement are folded into their supergenres, while making sure that all supergenres in the tree also meet the minimum song count. Again, as a probabilistic song count we employed the sum of assignment probabilities over all tracks.

We evaluated different minimum song counts using a grid search approach to find the optimal configuration that maximizes distribution uniformity within hierarchy levels, measured by the weighted within-level Gini coefficient. We calculated the algorithm for all possible solutions that branch at least once, starting with a minimum n of 1000. Given a range for the desired number of supergenres, a folding solution can be selected by taking the solution that corresponds to the local minimum of the weighted Gini within the given range. Additionally, local minima of the Gini over the number of supergenres function suggest the statistically best solutions the tree folding has to offer. For checking basic external validity of the solution, we build a contingency table comparing genre assignments on the Discogs folksonomy to a similar-sized solution of the COMGET folding. To facilitate this, we require songs in POPTRAG to have a single label. To assign single labels to the tracks from COMGET we employ a majority voting strategy: First we map every valid Musicbrainz genre tag that was assigned to a song to its supergenre in the folded COMGET. Then the track is assigned the supergenre tag with the highest number of summed votes. As a tie breaker, we choose the genre that has the lower sum of probabilities in the complete data set to promote a more more detailed genre assignments.

3.2 Results: Germany's Popular Music Supergenres

TODO: GINIPLOT

We applied the folding algorithm described above to COMGET and identified 4 optimal solutions (see figure X): a low detailed solution with 5 genres (min. 12,930 songs, Gini 0.307), a medium detailed solution with 12 genres (min 3,520 songs per genre, Gini 0.342), a high detailed solution with 31 genres (min 1,260 songs per genre, Gini 0.387) and a solution with even higher degree of detail comprising 38 supergenres (min 1,045 genres, Gini 0.385). The imbalance in-between the hierarchies increases until it reaches a sort of plateau from 20 supergenres upwards. Naturally the differences in minimum n between solutions with a lower number so supergenres is way higher than for solutions with a higher number of metagenres.

TODO: Metagenre graph with detail color code

The 5 genres solution comprises the supergenres *rock*, *pop*, *hip hop*, *electronic*, and *metal* (in descending order of their song counts, see figure X). For the medium resolution solution, *jazz* and *schlager* appear as distinct subgenres within the *pop* domain. All other tracks still remain mapped to the general *pop* category. Also, *heavy metal*, *alternative rock*, *pop rock*, *hard rock*, and *indie rock* appear as distinguished subgenres of *rock*. Notably, the majority of people do not see *heavy metal* as a subgenre of *metal*, but as a direct subgenre of *rock*, on the same hierarchy level as *metal*. The solution with 31 genres shows further distinct rock subgenres *punk*, *progressive rock*, *folk*, *folk rock*, *classic rock*, *blues rock*, *singer-songwriter*, *blues*, *country*, *new wave*, *soft rock*. Furthermore, a third level is introduced to the hierarchy with *death metal* as a subgenre of *metal* and *power metal* as subgenre of *heavy metal*. Within *pop*, *soul*, *classical*, *synth-pop*, *r&b*, *dance-pop*, and *reggae* appear as distinct subgenres. *hip hop* subgenres (*pop rap*, *trap*) do only appear in the very detailed solution with 38 gernes. In this very detailed solution, also electronic gets a subgenre (*house*), but overall no additional level remains in the resulting hierarchy.

TODO: COMGET vs Discogs plot

Discogs' genre taxonomy comprises 15 genres which is close to our 12 genres solution. The contingency table (figure X) shows a comparison of genre assignments between this solution (COMGET-12) and the corresponding assignments on *Discogs*. For ≈5% of tracks that have COMGET genre assignments, we did not find a *Discogs* assignment, introducing minimal popularity bias. We can see, that the assignments are unambiguous in general, especially with *electronic*, *hard rock*, and *metal*. Even with a *jazz* category present in the *Discogs* taxonomy, *jazz* in COMGET also contains some tracks assigned to *electronic* and *pop* on *Discogs*. *pop rock* mostly overlaps with *rock*, but also shows some slight overlap with *pop* and *electronic*. *schlager* mostly stems from *pop* (which corresponds to its COMGET supergenre *pop*) and shows some overlap with *rock*. Notably, *Discogs* genres *blues*, *brass & military*, *children's*, *non-music*, and *stage & screen* show no to little correspondence to any of the COMGET-12 genres. This makes sense in the light of the noise reduction (filtering out childrens' tracks and non-music tracks) and how the POPTRAG corpus was generated as a representation of popular music in Germany. The COMGET-12 genre *pop* is distributed across a wide array of *Discogs* Genres (most important *electronic*, *pop*, *rock*). This corresponds to the known eclectic nature of the *pop* label as described in the introduction. Finally, seven out of twelve genre in COMGET-12 have their greatest overlap with *Discogs*' *rock*. This indicates that the *Discogs* taxonomy lacks more detailed subgenre labels for rock genres that would be necessary to describe popular music in Germany in the past 50 years in a well-balanced way.

4. RQ3: ML-Classifier

4.1 Methods: Outcome, Features, Workflow

To determine wether an automated classifier can predict the supergenres that were identified using the folding algorithm presented in section 3, we first again derived single genre tags for each track in the data set using the majority vote (like we did for comparing COMGET-12 with the *Discogs* taxonomy). We did this for three different levels of detail (COMGET-5, COMGET-12, COMGET-31) and trained independent classifier for each level of detail. To collect track features on style, release milieu, lyrics, and popularity, we enriched POPTRAG with XXX variables from *Spotify*, *MusicBrainz*, *AcousticBrainz*, *Genius*, and *Deezer* using a fuzzy data linkage and webscraping. *AcousticBrainz* contains aggregated audio features that were calculated by the community on the full audio. *MusicBrainz* was mainly a source for information on artist persona, origin, birthyear, and gender. *Spotify* and *Deezer* provided further information on popularity, a tracks corresponding release, and echo nest audio features. From *Spotify*'s highly detailed multi-tagging artist genre system, we derived XX distribution genre flags (e.g., was this artist distributed as "alternative rock" artist) using the same algorithm we used to devise the COMGET tree, but with *Spotify* genre labels on artist level as input.

First, we tried different learners with a 20% subsample of the whole data set for low and medium resolution (high would not allow the upsampling algorithm) to determine which learner is the most suitable for our task. We tested elastic net classification (GLMNET),regularized discriminant analysis (RDA), random forest (RF), and light gradient boosted machine (LightGBM) to find a good compromise between prediction performance and model interpretability. For GLMNET we tuned penalty and mixture. For RDA we tuned the fraction of the common covariance matrix and fraction of the identity matrix. For RF we tuned the number of variables considered at each split (MTRY), minimum leaf node size (NMIN), and maximum depth of trees (MAXDEPTH, 500 trees were used during the tuning phase, 1000 trees for the final model). For GBM we tuned the number of boosting rounds (i.e., the number of trees), the learning rate, the number of tree leafs, maximum depth of trees, and the fraction of features used in each tree.

In our modelling workflow, we first discarded any observations with more than 40% missing values, retaining XX tracks (X%). We performed a train-holdout split (80%/20%) on the artist level to mitigate artist effects (Flexer and Schnitzer, 2010). We tuned our models with 5-fold cross-validation (also on artist level) and lumped all nominal levels in categorical predictors, that showed less than 100 obervations in any of the resulting folds, into an other category. Variables with only the other label remaining after this procedure were discarded. All categorical predictors were then dummy-coded with the most common level as reference category. We tuned the hyperparameters of our models on the $F1_{\text{macro}}$ score (mean of per-class F1 score, for classes that were not predicted F1 was set to 0) in order to have every category be treated as equally important when it comes to model evaluation.

During tuning, training, and evaluation, preprocessing steps were applied in sequence: median imputation, adaptive class balancing (not applied to holdout set), removal of zero-variance predictors, and normalization. Adaptive balancing employed a target-ratio parameter (tunable range: 1 to observed max:min ratio) that governed the degree of class rebalancing. The algorithm computed a reference count from class frequencies (median for ≤10 classes; 60th percentile for 11-20 classes; 65th percentile for >20 classes), then downsampled classes exceeding reference×target ratio using NearMiss (Zhang and Mani, 2003) and upsampled classes below reference÷target ratio using Borderline SMOTE (Han et al., 2005), and regular SMOTE in case of non sufficient danger observations. This approach preserved more data than fixed-ratio sampling while maintaining tunability: target ratio=1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic adjusted to problem complexity, preventing excessive downsampling in high-resolution scenarios.

We used a Bayesian optimization approach (Snoek et al., 2012) with a Latin hypercube initial grid (Husslage et al., 2011) with 20 grid points (for prototyping

10). We employed a maximum of 50 Bayes steps (for prototyping 25), set an early stopping criterion if no improvements occur after 10 iterations, and let the algorithm explore maximally uncertain areas of the hyperparameter space, if there were no improvements after 5 iterations. After selecting the hyperparameters with the one standard error rule (Han et al., 2005), we trained the final models on the full training set and evaluated the estimated models on the hold-out set. All random operations were seeded with "42".

4.2 Results: Performance in the Light of Uncertainty

TODO: PROTOTYPE LEARNERS COMPARISON PLOT

Model comparison (see fig x) shows that all four learners (GLMNET, RDA, RF, GBM) perform nearly equally well in terms of macro F1 on 20% of the data set. That is why we chose GLMNET to train models on the full data set due to its high interpretability, computational efficiency, and parsimontiy.

TODO: 3x CF GRAPHS

TODO: The final models on the full data set showed some underfitting. For classifier for the 5 genre solution yielded a mean $F1_{macro}$ of XX on the CV folds and $F1_{macro}$ of XX on the hold-out set; $F1$ per class highly positively correlated with XX with the assignment certainty within the community (= mean assignment probability for the class across tags). In the light of the mean assignment probability of XX, we can consider the classifier as very satisfactory. TODO: Description of the confusion matrix.

TODO: The classifier for the 12 genre solution also shows a little underfitting with mean $F1_{macro}$ of XX on CV folds and $F1_{macro}$ of XX on the holdout-set. Compare to the classifier for the low detailed solution classification performance was degraded. However, also the assignment certainty (XX%) was lower. Here, again F1 per class showed high correlation with per class voting certainty (XX). TODO:Description of the confusion matrix.

Finally, the classifier for the 31 genre solution again showed a degradation in performance with mean $F1_{macro}$ of XX on CV folds and $F1_{macro}$ of X on the hold-out set, matching a degradation in class assignment certainty within the community (XX%). Correlation of per class assignment certainty and is a little lower than it was for the other two levels of detail (XX).

TODO: Variable Importance Plot

TODO: A highly detailed discussion of the models coefficients/variable importance is not in the scope of this article (please refer to the digital appendix for a complete list of model coefficients/variable importance). Generally, it seems that artist features play a very important role in assigning genres from the listeners' perspective. For many genres, the marketing genres derived from the Spotify artist genres show to be most important, even if genre boundaries do not always align with the COMGET genre boundaries (e.g., the distribution genre *blues rock* was indicative for both *hard rock* and *rock* in the 12-genres-classifier). Among the audio features, *Spotify's* echo nest features TODO SOURCE prove to be most helpful for the classifier. Apart from the tracks language, the lyrics features that were used were not very important to the model.

5. Discussion

The aim of our study was to provide a method to devise a taxonomy of popular music genres. The resulting taxonomy should be well-balanced within hierarchical groups, scalable for different applications, and a machine learning algorithm should be able to predict genre tags with adequate performance for all genres, so that the taxonomy can be used with large corpora. We showed that it is possible by creating COMGET, a plausible and balanced taxonomy from genre tag co-occurrence in the MusicBrainz folksonomy that is based on popular music listened to in Germany during the past 50 years, captivated in the POPTRAG corpus. We introduced a folding algorithm that is able to scale this taxonomy to the desired number of supergenres, while suggesting statistically favorable solutions at different scales. Finally, we showed that it is possible to train an interpretable automatic classifier with satisfactory performance in line with the assignment uncertainty within the community at different levels of detail.

5.1 Balanced and Scalable Representation of Discourse

For our algorithm to construct a hierarchical genre taxonomy we adapted a simple and transparent algorithm proposed by Schreiber Schreiber (2015). However, for the source of our genre labels we avoided to mix commercial taxonomies with folksonomy. This is motivated by the power-critical viewpoint of Cultural Studies (du Gay et al., 2013), which is often adapted in popular music studies. This connects our method not only to previous studies in MIR, but also with debates within the humanities. Furthermore, our approach is consequently democratic: Not only do genre lables and their co-occurrence rely on the votes in a community of listeners, but democratic processes are deeply embedded within the proposed methodology: We used the strongest hierarchical link in search for the most plausible supergenre for each genre, and we also employed a majority voting with mapped gerne tags to convert the multi-label tagging system into single-label tags for training of the automated classifier.

Furthermore, our approach adheres directly to requirements for the use of genres in empirical environments. We propose a method to generate a balanced hierarchy without the need for manual thresholds or hyperparameters – portable to any corpus of musical works that has multi-tag genres available. We already showcased this possibility when inferring distribution genres from Spotify artist tags using POPTRAG. Also, our algorithm proposes optimally balanced solutions at different scales based on the same statistical criterion that was used to balance the full taxonomy. With this scaling algorithm it is possible to work on various research questions that operate on different levels of detail while maintaining a consistent genre framework. The method might, however, not only be of interest to researchers, but also to industry stakeholders like e.g., radio stations, or music streaming services, that need to organize large music collections at different levels of detail.

Crucially, the COMGET taxonomy does not strictly show genealogical links in the sense of historical developments as they were described by Pachet et al. Pachet et al. (2000). The taxonomy rather depicts a representation of the way the people talk about German popular music genres, i.e. how genres within music that is popular in Germany are related to each other in hierarchical discourse. Within this hierarchy, genealogical links like *rock* as supergenre of *alternative rock* are possible. However, a counterexample might be that *classical* and *jazz* appear as subgenres of *pop*. *classical* and *jazz* tracks in POPTRAG mostly comprise tracks that are accessible and exceptionally popular examples, e.g. from crossover artists, popular opera arias, soundtracks, popular jazz singers, or smooth jazz. Together with the very frequent use of broad labels (Bogdanov et al., 2019) this leads to the observed hierarchy. Also, *heavy metal* and *metal* are both direct subgenres of the *rock* genre. This can be explained by strong ties of the *heavy metal* to their roots in the psychedelic and hard rock music of the late 70s and the distinctive *metal* genres that appeared later (Gracyk, 2016).

We observed that rock, hip hop, and pop emerged as main supergenres in German popular music. These findings coincide with Silver et al.'s Silver et al. (2016) observation of three main genre clusters within international popular music. They investigated a large international sample of nearly 3 million MySpace artists self-selected genres, collected in 2007. In line with their observations, hip hop singles out as a community within popular music that does not strongly differentiate in subgenres, as can be inferred from the flat hierarchy of hip hop subgenres, their strong connection to the hip hop supergenre, as well as the small number of subgenres compared to the other two main genre clusters. However, while Silver et al. Silver et al. (2016) distinguished between "Rock' n Roll" and "Niche" genres, we find that rock genre form a distinct cluster in popular German music, that pop genres are diverse in terms of musical styles and that instead each cluster may have its own niche genres.

5.2 Classifier Performance in MGR

As Holt (Holt, 2007, p. 5) noted, semantic codes of musical genres are fragile. This makes MGR such a challenging endeavor. Even experts in the field do not entirely agree on genre assignments (P'almason et al., 2018); meaning that 'misclassifications' also reflect genre overlap. In other words: Models cannot exceed human consistency (Sturm and Flexer, 2023). That said, in the light of uncertainty in genre assignments within the community, the classifiers we trained showed reasonable overall performance. In line with (Genussov and Cohen, 2010) we also observe degrading classifier performance with increasing granularity of the genre labels. This goes along with a decreasing overall certainty for the community's genre assignments. Also, subgenres are harder to be detected than supergenres, probably due to shifting boundaries (Silver et al., 2016). We also observed different boundaries between genres that were assigned to market artists and genres that were assigned by the community of listeners, pointing to the intricate relationship between industry and listeners regarding the classification of music (Lena and Peterson, 2008; Negus, 1999), that should motivate a critical reflection when thinking about what genre taxonomies actually represent. In our models, the most important features were features on the artist level, indicating that release milieu is the most important features dimension to determine genres from the perspective of listeners. This represents a rather sociological understand of genres (Neale, 1980). However, stylistic features did also contribute significantly to the model. Features based on track lyrics did not show to be very important in modelling. However, lyric features have shown to be a helpful indicator in other studies (Mayer et al., 2008). It might therefore be beneficial to add LLM-based lyrics features to capture cultural context and improve discrimination of release milieus (Prevedello et al., 2024).

5.3 Limitations and Future Research

Our study faced several data and methodological limits that constrain generalization. MusicBrainz launched in 2000 and individual vote timestamps were unavailable, so the COMGET reflects listeners' genre votes aggregated over roughly the past 25 years rather than a single contemporary snapshot; genre boundaries and listener perceptions likely shifted over that period. The platform's user base and voting behavior remain opaque (≈1.2 million anonymous editor accounts, TODO https://musicbrainz.org/statistics), which prevented us from assessing demographic or expertise biases. Popularity bias also affected coverage: 36% of initial tracks received fewer than two MusicBrainz votes, so rare or niche recordings may have been underrepresented in the tree. However, as we want a taxonomy of Germanys popular music genres, this comes as only as a minor limitation. Methodologically, collinearity among metric predictors complicates interpretation of variable-importance scores and may have caused instability in which features appeared predictive across models.

Future work should validate and extend our findings along three lines. First, convene popular-music experts to review the taxonomy, label exemplar tracks, and flag misassignments to tighten COMGET's validity. Second, replicate the pipeline on independent folksonomies or regional MusicBrainz subsets (for example, US and UK listening histories) to test how national listening cultures reshape genre branches and to quantify cross-country differences. Third, refine classifiers using advanced lyrics features or learners.

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